AI Image Generator

5. Module Reference

Version 2.0
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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Z-Image Turbo, Stable Diffusion XL and its best fine-tunes, FLUX.1, LoRAs and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and a live Admin panel.

5. Module Reference

5.1 app.py (886 lines)

Entry point of the web application.

Name	Kind	Description
STARTUP_MODEL, OUTPUT_DIR, DEFAULT_NEGATIVE, UPSCALE_CHOICES, HELP	constants	Model loaded at start (zimage), outputs path, default negative prompt, the 2K/4K/8K choices, and the prompt help table.
PINK, YELLOW, GRADIENT, STATUS, CSS	constants	Brand colour ramps, the accent gradient, the four reserved status colours with icons, and the stylesheet for header, tiles and meters.
build_theme()	function	A Gradio Base theme with the Frazile AI palette applied to light and dark mode alike; only variables the installed Gradio accepts are set.
header_html(static_gpu)	function	The branded header with GPU, CUDA, torch and diffusers pills.
models_markdown()	function	The Models accordion: every model, defaults, on-disk state, plus LoRA and FLUX instructions.
MANAGER, UPSCALER, STATIC_GPU, MODEL_CHOICES	globals	The ModelManager (with sdxl loaded), the Upscaler, static GPU facts, and the dropdown choices, created once at import.
pick_model(settings, choice)	function	Resolves the model: prompt directive, then dropdown, then style-based auto choice limited to downloaded models. Returns (spec, note).
preview(prompt, choice)	function	The live "Will generate" line including the model and notes.
details_markdown(rec)	function	Renders one database record as the Markdown report shown under the gallery and in History.
upscale_record(rec, target_px, exact_size)	function	Upscales a record's native PNG, saves it, returns the eight upscale fields.
generate(prompt, negative, model_choice, progress)	function	Runs the full pipeline for all requested images. Returns gallery paths, report, records state, selection, Upscale bar visibility, hint and loaded line.
clear_all(), upscale_hint(), on_select(), upscale()	functions	Clear button, the hint under the Upscale bar, gallery selection tracking, and the Upscale bar handler that updates rows with db.update().
history_table(), history_stats(), refresh_history(), select_history()	functions	History tab, unchanged in behaviour but with a model column.
tile(), _kv(), _throttle_text(), gpu_energy_estimate(), admin_refresh(check_internet)	functions	Admin panel rendering: HTML stat tiles with meters and status, key-value tables, decoded throttle bits, the GPU energy estimate, and the collector that returns all seven outputs.

The UI is a gr.Blocks with fill_width=True and three tabs. Events: prompt change and model change to preview; Generate click and Enter to generate; Clear; gallery select to on_select; Upscale click to upscale; History tab select, Refresh and search to refresh_history; row select to select_history; Admin tab select, Refresh and a 5 s gr.Timer to admin_refresh, with the auto-refresh checkbox toggling the timer. launch() reads GRADIO_SERVER_NAME, GRADIO_SERVER_PORT and FRAZILE_NO_BROWSER and passes the theme and CSS.

5.2 models.py (436 lines)

Name	Kind	Description
ModelSpec	dataclass	key, name, source (HF repo or path), family (sdxl or flux), steps, guidance, scheduler, variant, negative, aliases, styles, description, download_gb, local.
BUILTIN	constant	The eight built-in specs with their defaults and recommended samplers.
local_checkpoints(), all_models(), model_words()	functions	Scan models/checkpoints/, the full list, and every key and alias for the parser.

Name	Kind	Description
resolve(name)	function	Finds a model by key, alias, name, repo id or path; raises KeyError if unknown.
model_for_style(style), is_downloaded(spec), display_name(source)	functions	Auto-selection lookup, on-disk check against the HF cache, and source-to-name mapping for the UI.
find_lora(name), available_loras()	functions	Case- and punctuation-insensitive lookup in models/loras/.
a1111_to_compel(prompt)	function	Rewrites (word:1.3) into compel's (word)1.3.
ModelManager	class	Holds one pipeline. load() dispatches to the SDXL, Z-Image or FLUX loader and reports progress; unload() frees VRAM; apply_loras() loads once and activates with weights; encode() returns weighted embeddings via compel with chunk padding, falling back to plain text; generation_kwargs() adds FLUX-specific arguments.

Running the module directly lists every model and whether it is downloaded.

5.3 prompt_parser.py (315 lines)

Name	Kind	Description
Settings	dataclass	prompt, width, height, steps, guidance, count, seed, style, upscale_to, exact_size, notes, model, loras, steps_explicit, guidance_explicit, quality, fidelity. Methods: summary(), upscaled_size(), apply_defaults(base_steps, base_guidance).
RATIO_BUCKETS, ORIENTATION_WORDS, STYLE_SUFFIXES, UPSCALE_TARGETS	constants	As in v1.0: eleven SDXL buckets, orientation regexes, twelve style presets, upscale words.
parse(text)	function	Returns a Settings. Order: LoRA tags, model directive, explicit resolution, ratio, orientation, upscale words, steps, guidance, count, seed, style, tidy-up, then apply_defaults with the SDXL baseline.

Quality and fidelity words no longer set fixed numbers; they are recorded and applied relative to the chosen model's defaults. Running the module directly prints the interpretation of twelve sample prompts.

5.4 sysinfo.py (358 lines)

Name	Kind	Description
human_bytes(), human_duration()	functions	Formatting helpers used by the Admin panel.
cpu()	function	Processor name (registry on Windows, /proc/cpuinfo on Linux), cores, frequency, usage per core, load averages, process and thread counts (cached 30 s), boot time, uptime.
memory(), disks()	functions	RAM and swap; partitions with usage, disk I/O totals, and the sizes of outputs/, models/, the HF cache and the database (cached 60 s).
network(), internet()	functions	Hostname, primary IP, totals, per-second rates from the previous sample, errors and drops, connections and listening ports, every interface; optional TCP reachability of three hosts.
gpu(), power()	functions	gpu_info static and live data plus PCIe link, VBIOS, max clocks, power limits, throttle reasons and GPU processes from nvidia-smi; power draw against limit, Windows power plan, battery.
software(), app(), everything()	functions	Library and driver versions and the Docker flag; the app process; one call that collects all of the above.

5.5 upscaler.py (109 lines)

Name	Kind	Description
MODEL_PATH, TARGETS, TILE, OVERLAP	constants	models/RealESRGAN_x4plus.pth; 2k 2560, 4k 3840, uhd 3840, 8k 7680; 512 and 32 pixels.
Upscaler(device, model_path)	class	Loads the model with spandrel, switches it to half precision on CUDA, exposes name and scale.
Upscaler.upscale_once(img)	method	One x4 pass over padded tiles under inference_mode.
Upscaler.upscale_to(img, long_side)	method	Repeats passes until the long side reaches the target, resizing before a second pass, then Lanczos-resamples to exactly long_side. Returns (image, info).

5.6 gpu_info.py (97 lines)

Name	Kind	Description
snapshot()	function	Runs nvidia-smi once and returns temperature_c, power_w, power_limit_w, gpu_util_pct, mem_util_pct, mem_used_mb, mem_total_mb, sm_clock_mhz, mem_clock_mhz, fan_pct, driver_version, pstate. Returns {} if nvidia-smi fails.
static_info()	function	gpu_name, vram_total_gb, compute_capability, multiprocessors, cuda_version, cudnn_version, torch_version, diffusers_version, python_version, os.
torch_memory()	function	vram_allocated_gb, vram_reserved_gb, vram_peak_gb from the torch allocator.

5.7 db.py (188 lines)

Name	Kind	Description
DB_PATH, SCHEMA, NEW_COLUMNS	constants	generations.db next to the module; CREATE TABLE plus index; columns added after the first release.
connect()	function	Opens a connection with Row factory, applies schema and migrations.
insert(record)	function	Inserts known keys as columns; unknown keys are packed into extra_json. Returns the row id.
update(row_id, fields)	function	New in v2.0: sets known columns on one row; used by the Upscale bar.
fetch(limit, search), get(row_id), stats()	functions	Newest-first list with LIKE search; one record; aggregates.

5.8 generate.py (121 lines)

Standalone command-line generator. Arguments: prompt, --negative, --steps, --guidance, --width, --height, --seed, --count, --model (key, repo id or path), --lora NAME[:WEIGHT] (repeatable), --out, --list-models. Uses the same ModelManager, so weighting, samplers and LoRAs behave exactly as in the app. It does not write to the database.

5.9 Supporting files

File	Purpose
requirements.txt	Python dependencies with the note on installing the CUDA PyTorch build first
run-app.bat	Double-click launcher that runs app.py inside the virtual environment
README.md	User-facing quick start, model table, prompt reference, Admin and Docker sections
Dockerfile, docker-compose.yml, .dockerignore	Container image and service definition (section 11)
models/RealESRGAN_x4plus.pth	Upscaler weights
models/checkpoints/, models/loras/	Drop folders for user SDXL checkpoints and LoRAs

File	Purpose
docs/	This documentation set as PDFs
docs-src/build_pdf.py, docs-src/images/	Source of this documentation and its figures